ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Adolescence: A Stage of Growth and Change** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** puberty	Puberty marks the beginning of reproductive maturity.
2	**(d)** puberty	Reproductive maturity.
3	**(b)** growth spurt	Sudden increase.
4	**(c)** 11 to 19 years	Teenage years.
5	**(d)** voice box (larynx)	Larynx enlarges.
6	**(b)** girls	Girls begin earlier.
7	**(a)** teenage	Ages 13 to 19 are teen years.
8	**(c)** Adam's apple	A lump in the throat.
9	**(d)** bathe and keep clean	Hygiene.
10	**(c)** hormones	Hormones travel in blood.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Period when body becomes sexually mature (1); About 10 to 12 years (1).	1+1
12	Monthly discharge of blood from the uterus (1); It occurs if the egg is not fertilised (1).	1+1
13	Sweat glands are more active (1); To avoid infections and body odour (1).	1+1
14	Growth of facial hair (1); Deepening of voice (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Rapid increase in height (1); Voice change (1); Growth of body hair (1).	1+1+1
16	Secretes adrenaline (1); Helps in emergencies (1); Increases heartbeat (1).	1+1+1
17	Healthy bones and muscles (1); Good mood (1); Controls weight (1).	1+1+1
	OR: Master gland (1); Controls other glands (1); Releases growth hormone (1).	1+1+1
18	Balanced diet (1); Exercise (1); Adequate sleep (1).	1+1+1
	OR: Body grows during sleep (1); Improves memory (1); Reduces stress (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Growth spurt (1); Voice change in boys (1); Facial hair in boys (1); Breast development in girls (1); Menstruation in girls (1).	1+1+1+1+1
	OR: Growth hormone (1); Thyroxine: goitre (1); Insulin: diabetes (1); Imbalance causes problems (1); Medical help (1).	1+1+1+1+1
20	Pituitary: growth hormone (1); Thyroid: thyroxine (1); Pancreas: insulin (1); Adrenal: adrenaline (1); Testes and ovaries: sex hormones (1).	1+1+1+1+1
	OR: Proteins for muscles (1); Calcium for bones (1); Vitamins for protection (1); Exercise strengthens (1); Prevents obesity (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that puberty starts at the same age in all children; it starts **earlier in girls** than in boys and varies from person to person.
- Writing that hormones are carried by ducts; endocrine glands are **ductless** and hormones travel in the blood.
- Confusing the glands: *testes* make testosterone, *ovaries* make oestrogen, *pancreas* makes insulin.
- Saying the voice breaks because of the throat or lungs; it is due to the enlargement of the **voice box (larynx)**.
- Calling menstruation an illness; it is a natural process and needs hygiene, not shame.
- Treating mood swings as bad behaviour; they are caused by hormones and can be managed with sleep, exercise and talking.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Adolescence: A Stage of Growth and Change** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** secondary sexual character	It appears at puberty.
2	**(c)** oil glands becoming more active	Hormones increase oil.
3	**(a)** endocrine glands	Glands without ducts.
4	**(b)** development of breasts	It appears at puberty.
5	**(b)** a balanced diet	Needs more nutrients.
6	**(b)** pituitary gland	It controls other glands.
7	**(c)** menstruation	Periods.
8	**(a)** iron	Menstruation causes blood loss.
9	**(b)** pituitary gland	Growth hormone.
10	**(c)** menarche	It starts in girls.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Growth is rapid (1); Body needs more nutrients (1).	1+1
12	Controls blood sugar (1); Made by the pancreas (1).	1+1
13	Development of breasts (1); Widening of hips (1).	1+1
14	Testes: testosterone (1); Ovaries: oestrogen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Menstruation causes blood loss (1); Iron makes haemoglobin (1); Prevents anaemia (1).	1+1+1
16	Controls metabolism (1); Needs iodine (1); Deficiency causes goitre (1).	1+1+1
17	Released in blood (1); Travel to target organs (1); Cause changes (1).	1+1+1
	OR: Enlarged voice box (1); Seen in boys (1); Voice becomes deeper (1).	1+1+1
18	Harm health (1); Addiction (1); Affect studies and relationships (1).	1+1+1
	OR: Primary: reproductive organs (1); Secondary: appear at puberty (1); Examples: voice, beard (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	More nutrients (1); Iron for girls (1); Calcium for bones (1); Daily bath (1); Clean clothes (1).	1+1+1+1+1
	OR: Boys: beard (1); Boys: deep voice (1); Girls: breasts (1); Girls: widening hips (1); Caused by sex hormones (1).	1+1+1+1+1
20	Egg released (1); Uterus lining forms (1); Shed as blood (1); Cycle about 28 days (1); Use clean sanitary products (1).	1+1+1+1+1
	OR: Many changes (1); Misleading sources online (1); Parents and teachers (1); Doctors (1); Books (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that puberty starts at the same age in all children; it starts **earlier in girls** than in boys and varies from person to person.
- Writing that hormones are carried by ducts; endocrine glands are **ductless** and hormones travel in the blood.
- Confusing the glands: *testes* make testosterone, *ovaries* make oestrogen, *pancreas* makes insulin.
- Saying the voice breaks because of the throat or lungs; it is due to the enlargement of the **voice box (larynx)**.
- Calling menstruation an illness; it is a natural process and needs hygiene, not shame.
- Treating mood swings as bad behaviour; they are caused by hormones and can be managed with sleep, exercise and talking.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Adolescence: A Stage of Growth and Change** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** goitre	Swelling of thyroid gland.
2	**(d)** testosterone	Made in the testes.
3	**(b)** menopause	End of reproductive years.
4	**(c)** thyroid gland	It controls metabolism.
5	**(a)** oestrogen	Made in the ovaries.
6	**(c)** 28 to 30 days	Typical cycle length.
7	**(d)** pancreas	Controls blood sugar.
8	**(d)** testosterone	It controls male changes.
9	**(b)** stops	Uterus lining is retained.
10	**(a)** diabetes	High blood sugar.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Iodine is needed for thyroxine (1); Prevents goitre (1).	1+1
12	Chemical messengers (1); Thyroid and pituitary (1).	1+1
13	Oil glands become active (1); Hormones change (1).	1+1
14	Hormone changes (1); Many new situations (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Body odour (1); Skin infections (1); Pimples worsen (1).	1+1+1
16	Mood swings (1); Talk to parents (1); Do hobbies (1).	1+1+1
17	Testes make sperms (1); Ovaries make eggs (1); Both make hormones (1).	1+1+1
	OR: Egg released from ovary (1); Uterus lining thickens (1); If not fertilised, lining sheds as blood (1).	1+1+1
18	Influence of friends (1); Think and say no to harmful things (1); Seek advice from trusted adults (1).	1+1+1
	OR: Secretes insulin (1); Lack causes diabetes (1); Sugar rises (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mood swings (1); Curiosity (1); Need for independence (1); Support and talk (1); Counselling (1).	1+1+1+1+1
	OR: Thyroxine metabolism (1); Iodine needed (1); Insulin controls sugar (1); Avoid excess sugar (1); Regular checkups (1).	1+1+1+1+1
20	Harms body (1); Causes addiction (1); Affects mind (1); Say no to peers (1); Seek help (1).	1+1+1+1+1
	OR: Everyone grows at own pace (1); No need to compare (1); Avoid teasing (1); Builds confidence (1); Good friendships (1).	1+1+1+1+1

Mistakes to avoid:

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